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**CAMPUS-AWARE INTELLIGENT LOST AND FOUND
MANAGEMENT SYSTEM USING NATURAL LANGUAGE
PROCESSING**

A CAPSTONE PROJECT REPORT

Submitted in partial fulfilment of the requirements for the Course of

DSA0308 – NATURAL LANGUAGE PROCESSING

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

CSE – ARTIFICIAL INTELLIGENCE

Submitted by

S. MOHAMMAD ABRAR (192472185)

Under the Supervision of

DR. T. KUMARAGURUBARAN

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

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Chennai – 602105

BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Campus-Aware Intelligent Lost and Found Management System Using NLP**” has been carried out by A. Syed Thoufiq (192424211), S. Mohammad Abrar (192472185), B. Praneeth (192472222), and K. Subrahmanya Venkata Sai Manikanta (192472220) under the supervision of Dr. T. Kumaragurubaran and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech CSE–Artificial Intelligence program at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR	COURSE FACULTY / GUIDE
Dr. Devi Professor Department of CSE SIMATS Engineering	Dr. T. Kumaragurubaran Professor Department of CSE SIMATS Engineering

Saveetha School of Engineering
SIMATS

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai – 602105

DECLARATION

We, A. Syed Thoufiq (192424211), S. Mohammad Abrar (192472185), B. Praneeth (192472222), and K. Subrahmanya Venkata Sai Manikanta (192472220) of the Department of Computer Science Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “Campus-Aware Intelligent Lost and Found Management System Using NLP” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.

We further declare that the data, results, analysis, and findings presented in this report have not been knowingly fabricated or falsified. All software libraries, datasets, references, and external resources used in the project have been appropriately acknowledged.

Place: Chennai

Date: 08/09/26

Signature of the Students with Names

A. Syed Thoufiq (192424211)

S. Mohammad Abrar (192472185)

B. Praneeth (192472222)

K. Subrahmanya Venkata Sai Manikanta (192472220)

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INDIVIDUAL CONTRIBUTION STATEMENT

The project was completed as a team effort. Each member was assigned primary responsibility for one core NLP module, while all members contributed to system integration, testing, documentation, and presentation. The contribution model below records the module-specific technical work and supporting activities.

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
A. Syed Thoufiq (192424211)	Project planning; Module 1 – Campus Context & Item Information Extraction; item, location and date/time extraction.	NER pipeline development; entity extraction and campus-context identification.	Test item, location and time extraction using varied lost/found descriptions; validate extracted entities.	Ch. 1 – Introduction; Module 1 documentation ; integration and presentation support.	25%
S. Mohammad Abrar (192472185)	Module 2 – Syntactic Structure Analysis; grammatical structure and word-category analysis.	HMM-based POS tagging using NLTK; noun, verb, adjective, preposition and modifier identification.	Test POS-tagging output and verify grammatical categories for different report descriptions.	Ch. 2 – Literature Review; Module 2 documentation ; testing and presentation support.	25%

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
B. Praneeth (192472222)	Module 3 – Linguistic Feature Extraction; conversion of descriptions into comparable numerical features.	TF-IDF weighting; unigram, bigram and trigram extraction; feature-vector construction.	Validate feature generation and compare feature representations for sample lost/found reports.	Ch. 3 – Engineering Design; Module 3 documentation ; feature-analysis and presentation support.	25%
K. Subrahmanya Venkata Sai Manikanta (192472220)	Module 4 – Lost-and-Found Similarity Matching; comparison, scoring and match ranking.	Cosine-similarity computation; location-aware comparison; ranking and threshold-based recommendation.	Test similarity scores, ranking behaviour and match recommendations using sample report pairs.	Ch. 4 & Ch. 5 – Implementation, Results and Conclusion; Module 4 documentation ; final verification and presentation support.	25%
TOTAL					100%

ABSTRACT

Lost and found management on university campuses becomes increasingly difficult to handle manually as the volume of daily reports grows across hostels, libraries, canteens, and academic blocks. Traditional processes rely on notice boards, manual registers, or spreadsheets and have limited ability to connect a lost report with its corresponding found item when different reporters use different words. This project presents FindIt Campus, a Campus-Aware Intelligent Lost-and-Found Management System Using Natural Language Processing (NLP), which applies NLP and information-retrieval techniques to understand free-text item descriptions, extract campus-related information, and identify likely matches between lost and found reports. The system collects item category, free-text description, campus landmark, reporter details, date, and report status. Named Entity Recognition (NER) extracts attributes such as item name, colour, brand, and location. Hidden Markov Model (HMM)-based Part-of-Speech (POS) tagging identifies grammatical categories such as nouns, adjectives, and prepositional phrases. TF-IDF weighting combined with unigram, bigram, and trigram features converts descriptions into comparable numerical vectors. Cosine similarity is then computed between lost and found vectors, and candidate pairs above the configured similarity threshold of 0.70 are ranked and surfaced through the Find Matches action and administrative dashboard. Representative campus scenarios involving a backpack, headphones, and a student ID card showed that the correct counterpart was ranked first despite differences in wording. An unmatched charger case was also retained as open/unmatched, demonstrating that the system does not need to force a match. The project therefore demonstrates how lightweight, interpretable NLP can reduce manual reconciliation effort and support faster item recovery while retaining human verification before resolution.

Keywords: Lost and Found Management, Natural Language Processing, Named Entity Recognition, Part-of-Speech Tagging, TF-IDF, Cosine Similarity, Text Similarity Matching, Information Retrieval, Campus Management System, Smart Campus

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description	Unit
NLP	Natural Language Processing	—
NER	Named Entity Recognition	—
POS	Part-of-Speech	—
HMM	Hidden Markov Model	—
TF-IDF	Term Frequency – Inverse Document Frequency	—
API	Application Programming Interface	—
UI/UX	User Interface / User Experience	—
CRUD	Create, Read, Update, Delete	—
JSON	JavaScript Object Notation	—
DB	Database	—
N-gram	Contiguous sequence of N words	—
SDG	Sustainable Development Goal	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

University campuses are dense, high-traffic environments in which personal belongings are frequently misplaced. Backpacks may be left in canteens, headphones forgotten in library reading rooms, ID cards dropped near administrative buildings, and chargers abandoned in laboratories. Traditionally, institutions manage such cases using a physical lost-and-found counter, notice board, register, or shared spreadsheet. When reports increase, a staff member must manually inspect descriptions and try to connect a lost report with a found item.

The central difficulty is that users do not describe objects in a standardized way. One person may write “blue Wildcraft backpack near the CSE block canteen,” while another may write “navy blue Wildcraft laptop bag left at a table near the juice stall.” Humans can recognize the connection from the shared context, but exact keyword search may not. NLP provides a practical way to normalize, structure, and compare such free-text descriptions.

1.2 Need for the Project

1. Manual reconciliation becomes slow as the number of reports grows.
2. Different users use different nouns, adjectives, abbreviations, and levels of detail for the same object.
3. Exact keyword matching can miss valid matches when wording changes.
4. Campus location is an important contextual signal for matching.
5. A centralized platform can keep report status auditable and reduce repetitive staff effort.

1.3 Problem Statement

The engineering problem is to design and implement a web-based campus lost-and-found system that accepts free-text descriptions, extracts useful item and campus information, converts descriptions into comparable features, and automatically identifies and ranks likely matches between lost and found reports while keeping a human decision-maker in control of final resolution.

1.4 Project Aim

To design and develop an intelligent campus lost-and-found platform that uses lightweight NLP techniques to extract information from user descriptions and rank likely lost-and-found matches efficiently.

1.5 Project Objectives

1. Design a guided web platform through which students and staff can report lost and found items.
2. Extract structured information such as item, location, colour, brand, and time from free-text descriptions using NER.
3. Identify grammatical structure and useful word categories using HMM-based POS tagging.
4. Convert descriptions into numerical features using TF-IDF and unigram, bigram, and trigram representations.
5. Compute cosine similarity between lost and found descriptions and rank likely matches.
6. Provide an administrative view of open, matched, and resolved reports.
7. Evaluate the system using representative campus scenarios and check that genuine matches rank above unrelated reports.

1.6 Scope

The project covers reporting lost and found items, browsing and filtering open reports, automatic candidate-match generation, similarity ranking, and administrative oversight of the report lifecycle. The current implementation is text-description driven. Although the reporting forms allow an optional photograph, image-based visual matching is outside the current similarity engine and is treated as future work. The intended deployment boundary is a single campus with a defined set of locations and item categories.

1.7 Assumptions and Constraints

The current prototype assumes a defined set of campus locations and item categories, reasonably descriptive user input, and access to stored lost/found reports. The matching engine is intended for a single-campus deployment and relies on text descriptions; optional photographs are not used by the current similarity engine. Final ownership decisions remain

subject to human verification, and the system does not require a large labelled campus-specific training corpus.

1.8 Expected Engineering Outcome

The expected outcome is a maintainable and interpretable campus application in which classical NLP techniques operate as an intelligent matching layer inside a conventional reporting and database workflow. The system should reduce manual searching, surface likely matches quickly, and preserve human verification before an item is marked resolved.

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CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The project was informed by established textbooks and research in natural language processing, information retrieval, named entity recognition, POS tagging, term weighting, and hidden Markov models. The selected techniques were compared on suitability for short campus descriptions, computational cost, interpretability, and ease of integration into a web application.

2.2 Review of Existing Work

Conventional campus processes rely on static notices, registers, spreadsheets, or manual staff review. A second approach is exact keyword search, which is fast but brittle when descriptions use different words. More advanced approaches can use learned semantic embeddings or multimodal image-text models, but they generally require additional model infrastructure and, depending on the approach, domain data or tuning.

2.3 Comparative Analysis

Table 1 – Existing Approaches Comparison

Existing Approach	Advantage	Limitation	Relevance
Static register / notice board	Simple and familiar	No automatic matching; difficult to scale	Baseline manual process
Exact keyword search	Fast and easy to implement	Fails with vocabulary variation	Insufficient for the target problem
TF-IDF + N-grams + cosine similarity	Fast, interpretable, lightweight	Sensitive to vocabulary mismatch and deeper semantics	Selected for current project
Transformer-based semantic	Better semantic representation	Higher compute and implementation	Future enhancement

Existing Approach	Advantage	Limitation	Relevance
embeddings		complexity; domain tuning may be needed	
Image-based matching	Can use visual appearance	Requires image model and reliable photos	Future enhancement

2.4 Research / Engineering Gap

The project identifies a practical gap between manual campus registers and heavier AI systems: there is a need for a lightweight, explainable matching layer that can work on short free-text reports without requiring a large campus-specific training corpus. The proposed design addresses this gap by combining structured extraction, linguistic analysis, vector-space feature engineering, and similarity ranking.

2.5 Proposed Contribution

The contribution is an integrated FindIt Campus workflow that combines four NLP modules—Campus Context and Item Information Extraction, Syntactic Structure Analysis, Linguistic Feature Extraction, and Lost-and-Found Similarity Matching—with reporting forms, browsing/filtering, a database, and an administrative dashboard. The pipeline is deliberately modular so that each stage can be tested independently.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The **FindIt Campus** system is designed to provide a simple and intelligent lost-and-found management platform for university campuses. The requirements are identified based on the needs of students, faculty, campus staff, security personnel, and administrators. The system combines conventional web-based reporting with an NLP pipeline for extracting information and identifying possible matches.

3.2 Constraints & Applicable Standards

Table 3 – Functional and Non-Functional Requirements

Requirement	Type	Target
Report Lost / Found	Functional	Successful submission of required report fields
NLP extraction	Functional	Item/location/time and descriptive entities processed
Similarity matching	Functional	Candidate matches ranked using cosine similarity
Search and filters	Functional	Reports filterable by type, category, location, and status
Admin dashboard	Functional	Open / Matched / Resolved reports visible
Processing efficiency	Non-functional	Matching runs fast enough for report-submission workflow
Usability	Non-functional	Guided forms understandable to non-technical users

Table 2 – Stakeholder / User Requirements

Stakeholder	Requirement
Students / Faculty	Fast, simple reporting; view possible matches; track status
Security / Administration	Verify matches; manage report status; monitor dashboard
Library / Canteen / Department Staff	Quickly submit found-item reports without specialized training
Campus Administration	Monitor open cases, matching activity, and resolution trends

3.3 Design Specifications

- Maintainability: lightweight libraries and separated NLP stages make debugging and future replacement of a module easier.
- Auditability: report status follows an explicit lifecycle such as Open → Matched → Resolved.
- Consistent taxonomy: item categories and campus locations are standardized across reporting, search, and administration.
- Data validation: required reporter details, category, and description are checked before processing.
- Modular design: the four NLP capabilities have clear inputs and outputs and can be tested independently.

3.4 Alternative Solutions & Evaluation

Table 4 – Design Specifications

Design Parameter	Target	Priority
Matching quality	True matches should be ranked above unrelated reports	High
Processing time	Fast enough for report-submission workflow	Medium
Interpretability	Similarity results should be understandable to administrators	High
Usability	Guided reporting and clear match presentation	High

Design Parameter	Target	Priority
Privacy and security	Collect only necessary report/contact information and restrict access	High
Maintainability	Four NLP modules remain independently testable	Medium

3.5 Selected Design & Detailed Design

3.5.1 Module 1 – Campus Context & Item Information Extraction

Objective: extract important information such as item, location, and time from user-submitted lost/found descriptions. The module parses the item description and detects campus-specific context such as block, room, or floor. Named Entity Recognition (NER) is the main technique.

Example: “I lost my black Dell laptop in Block B yesterday.” The desired structured output includes item/laptop, colour/black, location/Block B, and time/yesterday. The structured attributes can later be used as auxiliary signals during matching.

3.5.2 Module 2 – Syntactic Structure Analysis

Objective: analyze the grammatical structure of the description and identify important word categories. HMM-based POS tagging is used to assign tags such as noun, verb, adjective, and preposition. The Hidden Markov Model considers transition probabilities between tags and emission probabilities connecting words to likely tags, then selects a probable sequence of tags.

Example: “I lost my black leather wallet near Block A.” The tagger identifies “lost” as a verb, “black” and “leather” as adjectives, “wallet” as a noun, and “near” as a preposition. This grammatical structure helps distinguish descriptive words from other parts of the sentence.

3.5.3 Module 3 – Linguistic Feature Extraction

Objective: convert descriptions into meaningful numerical features for comparison. TF-IDF assigns higher weight to terms that are distinctive within the report collection, while n-grams preserve short word sequences. The implementation uses unigram, bigram, and trigram features and builds a vector for each report.

For example, “navy blue Wildcraft backpack near CSE canteen” can produce unigram features such as “navy”, “blue”, and “backpack”, bigrams such as “navy blue” and “CSE canteen”, and trigrams such as “navy blue Wildcraft”. These features are combined into a comparable vector representation.

3.5.4 Module 4 – Lost-and-Found Similarity Matching

3.6 Trade-off Analysis

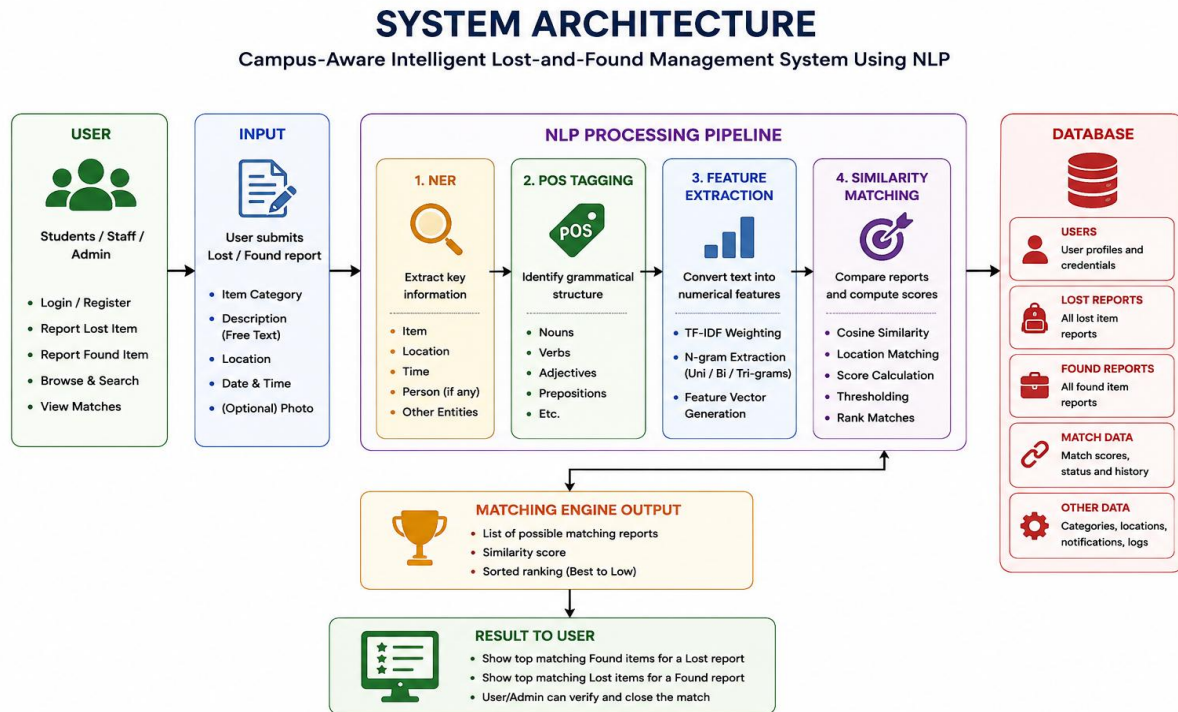


Figure 1 – System Architecture of FindIt Campus

The system follows a simple layered workflow: the user submits a lost or found report; the backend sends the free-text description through the NLP pipeline; the resulting features are compared with reports of the opposite type; candidate matches are ranked; and the results are returned to the user or administrator. Report data and match information are persisted in the database.

Table 6 – Trade-off Analysis

Design Option	Benefit	Trade-off	Decision
Exact keyword matching	Very low computation	Sensitive to vocabulary	Rejected as primary method

		variation; may miss valid matches	
TF-IDF + n-grams + cosine	Low compute, interpretable, easy to deploy	Primarily lexical; weaker deep semantic understanding	Selected
Transformer embeddings	Better semantic robustness	Higher compute and engineering complexity	Future enhancement
Image similarity	Adds visual evidence	Requires image model and reliable photographs	Future enhancement

3.7 Sustainability, Ethics, Safety, Risk & Security

A score closer to 1 indicates greater similarity between the vector representations. In the project example, “iPhone lost near Block C” is compared with a corresponding found report and a similarity score of 0.87 is used to illustrate a strong match. The system does not treat a score as proof of ownership; final confirmation remains with a human.

$$\cos(\theta) = (\mathbf{A} \cdot \mathbf{B}) / (||\mathbf{A}|| \times ||\mathbf{B}||)$$

Cosine similarity is calculated as:

Objective: compare lost and found reports and rank the most likely matching items. The system computes cosine similarity between the feature vectors of reports of opposite types. Location matching is also considered as a campus-context signal. Candidate pairs above the configured threshold of 0.70 are surfaced for review and ranked from highest to lowest score.

Table 5 – Alternative Solutions Evaluation

Approach	Strength	Trade-off	Decision
Exact keyword matching	Very low compute	Poor with vocabulary variation	Rejected as primary method
TF-IDF + n-grams +	Low compute;	Lexical rather than deep	Selected

Approach	Strength	Trade-off	Decision
cosine	interpretable; easy to deploy	semantic matching	
Transformer embeddings	Better semantic robustness	Higher compute and engineering complexity	Future enhancement
Image similarity	Useful visual signal	Requires image model and good photos	Future enhancement

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

Table 8 – Risk Register

Risk	Impact	Level	Mitigation	Residual Risk
False-positive match	Wrong item could be handed to an incorrect claimant	High	Human verification; threshold; status control	Medium
False-negative match	Valid pair may not be surfaced	Medium	N-grams; location signal; configurable threshold	Medium
Poor description	Insufficient matching signal	Medium	Guided form; category and location fields	Low/Medium
Unauthorized access	Exposure of report/contact information	High	Authentication/access control; data minimization	Low
Incorrect status update	Loss of report traceability	Medium	Controlled status transitions and logging	Low

Table 7 – Sustainability, Ethics, Safety and Security Evidence

Domain	Consideration	Evidence Approach	Impact on Final Decision
Sustainability	Reuse and waste reduction	NLP matching can help reunite owners with existing belongings; project aligns with SDG 12.	Supports reuse over unnecessary replacement.
Ethics	Responsible use of report and contact data	Data minimization, transparency and human-in-the-loop verification.	Reduces privacy and false-positive risks.
Safety	Safe software operation	Current system only performs reporting, NLP processing and matching; no physical control.	No direct physical safety hazard in the prototype.
Security	Protection of reports and personal details	Authentication/access control and limited collection of personal information.	Reduces unauthorized access risk.

The project supports SDG 12 by helping recover and reuse existing belongings instead of encouraging unnecessary replacement. Ethical safeguards include data minimization, transparency about NLP processing, access control, and human verification of candidate matches. The current prototype is software-based and does not create physical safety hazards; operational risk is concentrated in incorrect matches, poor descriptions, and unauthorized access.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project followed an incremental development approach covering problem analysis, requirements identification, NLP pipeline design, module implementation, application integration, testing, result evaluation, documentation, and presentation. Development resources included Python, NLTK, Scikit-learn, NumPy, a database-backed web application workflow, and existing computing resources. The approach emphasized modular development so that each NLP stage could be tested independently before integration.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

1. User selects Report Lost or Report Found and enters category, date, location, and free-text description.
2. Backend validates required fields and stores the report.
3. NER extracts item-related and campus-related attributes.
4. HMM-based POS tagging identifies grammatical categories.
5. TF-IDF and n-gram extraction produce a numerical feature vector.
6. The vector is compared against open reports of the opposite type using cosine similarity.
7. Location/context is checked and candidate pairs above the 0.70 threshold are ranked.
8. The Find Matches view and Admin dashboard display candidate matches for human verification.

4.2.1 Tools / Software / Technologies Used in FindIt Campus

Table 9 – Tools / Software / Technologies Used in FindIt Campus

Layer / Module	Technology / Technique	Purpose
Frontend UI	Responsive web application	Guided reporting, browse/filter directory, admin console
NER / item extraction	Python NLP libraries	Extract item, location, time, colour, brand
POS tagging	HMM-based POS tagging,	Grammatical categories and

Layer / Module	Technology / Technique	Purpose
	Python, NLTK	structure
Feature extraction	TF-IDF + unigram/bigram/trigram, Scikit-learn, NLTK	Numerical feature vectors
Similarity matching	Cosine similarity, Scikit-learn, NumPy	Similarity score and ranking
Data storage	Relational / document database	Lost/found reports, users, status, match data
Administration	Dashboard with filters and status controls	Open / Matched / Resolved monitoring

4.3 Test Plan & Verification

The verification plan checked report submission, entity extraction, syntactic tagging, feature generation, similarity ranking, unmatched-case handling, and the administrative report lifecycle.

Table 10 – Test Plan & Verification

Test Case	Input / Scenario	Expected Result	Outcome
Report submission	Valid lost/found form	Report saved successfully	Pass
NER extraction	Backpack/headphones/ID examples	Key item and campus entities extracted	Pass
POS tagging	Representative descriptions	Noun/adjective/verb/preposition tags produced	Pass
Feature generation	Known sample descriptions	TF-IDF and n-gram vectors created	Pass
True-match ranking	3 known matched pairs	Correct counterpart ranked first above 0.70	Pass
Unmatched	Lost 65W Dell USB-C	No forced found-item match;	Pass

Test Case	Input / Scenario	Expected Result	Outcome
case	charger	remains open	
Admin workflow	Open → Matched → Resolved	Status visible and controllable	Pass

4.4 Results

The project evaluation used representative campus scenarios covering Bags & Backpacks, Electronics, and ID Cards. Three known lost/found pairs were tested, together with an unmatched charger report. The correct counterpart was consistently ranked first for the three true-match cases, even when wording differed between the lost and found descriptions.

Table 11 – Matching Results / Key Outcomes

Pair	Category	Vocabulary Overlap	Result
Backpack (#st-101 / #nd-201)	Bags	Partial overlap; “Wildcraft” shared, colour/noun differ	Correctly ranked top match
Headphones (#st-102 / #nd-202)	Electronics	Partial overlap; “Sony” shared, model vs generic wording	Correctly ranked top match
ID Card (#st-103 / #nd-203)	ID Cards	High overlap; roll number and lanyard colour shared	Correctly ranked top match; status Matched
Charger (#st-104)	Electronics	No corresponding found report	Correctly left Open / Unmatched

4.5 Data Analysis & Engineering Judgment

During the development/test window, the platform included five lost reports and five found submissions across Bags & Backpacks, Electronics, and ID Cards, with reports associated with CSE Block Canteen, Central Library, Admin Building, and Science Block Lab. The administrative view recorded 10 active NLP-generated candidate pairs and 18 items resolved in the current week during the reported test snapshot. These values are operational observations from the project test environment, not a statistically representative campus-wide benchmark.

Table 12 – Module Performance Parameters

Module	Technique / Tooling	Key Parameters Monitored
Campus Context & Item Information Extraction	NER, Python NLP libraries	Entity type; extracted entity; extraction accuracy
Syntactic Structure Analysis	HMM-based POS tagging, NLTK	POS tag; word category; tagging accuracy
Linguistic Feature Extraction	TF-IDF + N-gram extraction, Scikit-learn, NLTK	TF-IDF score; n-gram frequency; feature vectors
Lost-and-Found Similarity Matching	Cosine similarity, Scikit-learn, NumPy	Similarity score; ranking; threshold = 0.70

Engineering judgment: The representative tests indicate that the selected classical NLP pipeline is suitable for the current single-campus scope. The strongest evidence is consistent top-ranking of the known backpack, headphones, and ID-card counterparts, while the unmatched charger remained open. These observations support retaining the lightweight pipeline with a configurable threshold and human verification, while recognizing that larger-scale evaluation is still required for broader deployment. Key challenges observed included vocabulary drift such as “backpack” versus “bag”, short-text sparsity, threshold selection, ambiguous descriptions, and the need for human verification because similarity is a ranking signal rather than proof of ownership.

Recommendation: The current text-based pipeline should remain the primary matching mechanism for the present scope because it is lightweight, interpretable, and performed well on the representative scenarios tested. The similarity threshold should remain configurable, and future visual matching and notifications should be introduced as additional signals rather than immediately replacing the current pipeline.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Compared with static registers and exact keyword search, FindIt Campus provides automated candidate generation and ranking while remaining more lightweight and interpretable than transformer-based semantic matching. The implemented objectives are supported by the four

NLP modules, the reporting workflow, administrative status controls, and representative true-match and unmatched test cases.

Table 13 – Objective Achievement

Objective	Evidence	Achievement
Provide guided lost/found reporting	Report Lost and Report Found workflows are implemented.	Achieved
Extract structured information	NER extracts item and campus-related attributes from descriptions.	Achieved
Analyse grammatical structure	HMM-based POS tagging is included in Module 2.	Achieved
Generate comparable text features	TF-IDF and unigram/bigram/trigram features are generated.	Achieved
Rank likely matches	Cosine similarity, location context and threshold-based ranking are implemented.	Achieved
Support administration	Admin view provides report status and matching oversight.	Achieved
Validate matching behaviour	Backpack, headphones and ID-card scenarios ranked the correct counterpart first; charger remained unmatched.	Achieved

4.7 Strengths & Limitations

Strengths: The system integrates four complementary NLP stages into one campus workflow, uses lightweight and interpretable techniques, incorporates campus context such as location, supports ranked candidate matching, and retains human verification before resolution. The modular architecture also makes individual components easier to test and replace.

Limitations: The current matching engine is primarily lexical and may be affected by vocabulary mismatch, very short descriptions, ambiguous item details, and threshold selection. The prototype is intended for a single-campus scope and does not currently use image-based matching or advanced semantic embeddings.

4.8 Project Planning, Roles & Budget

Project planning covered problem analysis, NLP design, application design, integration, testing, and final documentation. The project used equal team contribution across the four assigned NLP modules, with shared integration, testing, documentation, and presentation responsibilities.

Table 14 – Project Management Milestones

Milestone	Main Activity	Status
Problem analysis	Identify manual matching problem and user workflow	Completed
NLP design	Define NER, HMM-POS, TF-IDF/n-gram and cosine pipeline	Completed
Application design	Build reporting, browsing and admin workflow	Completed
Integration	Connect NLP pipeline with stored reports and Find Matches action	Completed
Testing	Run matched and unmatched campus scenarios	Completed
Documentation	Prepare capstone report and presentation	Completed

Table 15 – Project Roles

Student	Assigned Responsibility
A. Syed Thoufiq (192424211)	Module 1 – Campus Context & Item Information Extraction; project planning and integration support
S. Mohammad Abrar (192472185)	Module 2 – Syntactic Structure Analysis; literature review and testing support

B. Praneeth (192472222)	Module 3 – Linguistic Feature Extraction; feature analysis and testing support
K. Subrahmanya Venkata Sai Manikanta (192472220)	Module 4 – Lost-and-Found Similarity Matching; results, verification and presentation support

4.8.1 Project Budget

The project is primarily software-based and uses open-source NLP/Python libraries and existing development resources. No separate software or dataset expenditure was reported.

Table 16 – Project Budget

Item	Estimated Cost	Actual Cost
Python / NLP libraries	Open-source	No separate cost reported
Development environment	Existing resources	No separate cost reported
Computer / development system	Existing resources	No separate cost reported
Dataset / sample reports	Project data / sample reports	No separate cost reported
Printing / documentation	Not separately reported	Not separately reported
TOTAL	Not separately reported	Not separately reported

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

FindIt Campus demonstrates that a lightweight NLP pipeline can improve a practical campus workflow. The system accepts free-text lost and found descriptions, extracts useful entities, analyzes syntactic structure, creates TF-IDF and n-gram features, and ranks candidate matches using cosine similarity. Representative tests showed that the correct counterpart was ranked first for the tested backpack, headphones, and student ID card scenarios, while an unmatched charger remained open rather than being forced into a false match.

The project also demonstrates an important engineering principle: the most sophisticated model is not always the most suitable model. For the present single-campus scope, a transparent and computationally inexpensive pipeline is easier to explain, maintain, and integrate. The system supports faster reunions and encourages reuse of existing belongings, aligning with the broader intent of SDG 12 – Responsible Consumption and Production.

5.2 Future Work

1. Mobile application for easier reporting and tracking.
2. Advanced semantic AI techniques for improved vocabulary robustness.
3. Real-time notifications for potential matches.
4. QR-code based reporting and item handover workflows.
5. Campus map and location-aware features.
6. Multilingual reporting and matching.
7. Image-based similarity using uploaded item photographs.

5.3 Professional Learning & Individual Reflection

5.3.1 New Knowledge Acquired

The project strengthened understanding of natural language processing as a pipeline rather than a single algorithm. The team learned how NER extracts structured information,

how HMM-based POS tagging models sequential word categories, how TF-IDF and n-grams represent short text, and how cosine similarity compares vector representations.

5.3.2 Engineering & Professional Skills Developed

The work provided practical experience with Python, NLTK, Scikit-learn, NumPy, database-backed application workflows, form validation, text preprocessing, vectorization, similarity scoring, ranking, dashboard-oriented presentation, testing, documentation, and teamwork.

5.3.2.1 Problem-Solving and Critical Thinking

A key learning was that the hardest part of the problem is vocabulary variation in very short descriptions. The team therefore had to balance accuracy, computational cost, explainability, and implementation effort. The resulting classical NLP pipeline was chosen because it provides a practical solution for the current campus scale without requiring a large labelled training dataset.

5.3.2.2 Challenges Encountered and Overcome

The major challenges were vocabulary drift, short descriptions, threshold selection, and ambiguous item descriptions. These were addressed through n-gram features, auxiliary category/location information, empirical threshold testing, and human-in-the-loop verification. The process demonstrated the importance of testing both true-match and true-non-match cases.

5.3.2.3 Application of Engineering Practices

The project applied modularity, validation, consistent taxonomies, traceable report status, and separation between the NLP processing layer and the application interface. These practices improve maintainability and make individual stages easier to test and replace.

5.3.2.4 Application of Ethical Principles

Ethical design was treated as a core requirement because a false-positive match could result in personal property being given to the wrong person. The system therefore proposes matches rather than making the final ownership decision. Data minimization, transparency about NLP processing, access control, and human verification are important safeguards.

5.3.3 Individual Reflection – A. Syed Thoufiq (192424211)

My primary contribution was Module 1, Campus Context & Item Information Extraction. I worked on extracting item, location, colour, brand, and date/time information from free-text lost-and-found descriptions using NER-oriented processing. This work strengthened my understanding of information extraction and showed how structured entities can provide useful context for downstream matching. I also contributed to project planning, integration, documentation, testing, and presentation.

5.3.4 Individual Reflection – S. Mohammad Abrar (192472185)

My primary contribution was Module 2, Syntactic Structure Analysis. I worked on HMM-based POS tagging to identify nouns, verbs, adjectives, prepositions, and other useful grammatical categories in item descriptions. This improved my understanding of statistical sequence tagging and its practical role in NLP pipelines. I also contributed to literature review, testing, documentation, integration, and presentation activities.

5.3.5 Individual Reflection – B. Praneeth (192472222)

My primary contribution was Module 3, Linguistic Feature Extraction. I worked on TF-IDF weighting and unigram, bigram, and trigram feature extraction to convert item descriptions into comparable numerical representations. This helped me understand how lexical features influence text similarity and how feature engineering affects matching quality. I also contributed to feature validation, documentation, integration, and presentation.

5.3.6 Individual Reflection – K. Subrahmanya Venkata Sai Manikanta (192472220)

My primary contribution was Module 4, Lost-and-Found Similarity Matching. I worked on cosine-similarity scoring, location-aware comparison, ranking, and threshold-based candidate recommendation. This strengthened my understanding of vector-space similarity and the importance of evaluating both correct and unmatched cases. I also contributed to implementation, results analysis, final verification, documentation, and presentation.

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APPENDICES

Appendix A — System Screenshots and Module Reference

Figure A1 – Report Lost Item Interface

The screenshot displays the 'Report Lost' form in the FindIt Campus system. The interface is divided into several sections for data entry:

- Header:** Includes the FindIt Campus logo, a search bar, navigation links (Home, Browse All, Admin), and action buttons for 'Report Lost' and 'Report Found'. A user profile for Aarav Sharma is visible in the top right.
- Presets:** A section titled 'Faculty Viva / Fast Demo Presets (Click to autofill):' offers quick selection of items like '+ Blue Wildcraft Backpack (CSE)', '+ Sony WH-1000XM4 Headphones (Library)', and '+ Student ID Card (Admin Block)'.
- 1. REPORTER DETAILS:** Contains fields for 'Your Full Name' (filled with 'Aarav Sharma') and 'Contact Email or Phone' (filled with 'aarav.s@campus.edu').
- 2. ITEM SPECIFICATIONS & DESCRIPTION:** Includes an 'Item Category' dropdown (set to 'Bags & Backpacks'), a 'Date Lost' field (set to '08/28/2026'), and a 'Free-text Description' area. A tip below the description field states: 'Tip: Include specific attributes like color, brand, stickers, or distinguishing marks to maximize cosine similarity matches.'
- 3. CAMPUS LOCATION / LANDMARK:** Features a 'Campus Landmark' dropdown (set to 'CSE Block Canteen').
- 4. PHOTO (OPTIONAL):** A section for uploading a photo, with a note 'Stored client-side as base64'. It includes a drag-and-drop area and a file selection button.
- Footer:** A note states 'Submitting triggers automatic real-time vector matching against opposite reports.' and a 'Submit Lost Item Report' button.

Report Lost form showing reporter details, item specification, campus landmark, and description input.

Figure A2 – Report Found Item Interface

FindIt Campus
Intelligent Lost & Found System

Search items, tags, k Home Browse All Admin Report Lost Report Found 2 Aarav Sharma STUDENT

FOUND PROPERTY REGISTRY
Report a Found Item
Help reunite items with their rightful campus owners. Your report will be cross-referenced against all active lost listings.

Faculty Viva / Fast Demo Presets (Click to autofill):
+ Found Navy Wildcraft Bag (Canteen) + Found Sony Wireless Headphones (Library)
+ Found Student ID Card (Admin)

1. REPORTER DETAILS

Your Full Name * Contact Email or Phone *

Aarav Sharma aarav.s@campus.edu

2. ITEM SPECIFICATIONS & DESCRIPTION
The backend NLP parser will automatically extract named entities (brands, colors) and POS tokens from your description.

Item Category * Date Found

Bags & Backpacks 08/28/2026

Free-text Description * 0 chars (NLP vector input)

e.g. "Found a navy blue Wildcraft laptop bag left unattended at CSE canteen table near the juice stall."

Tip: Include specific attributes like color, brand, stickers, or distinguishing marks to maximize cosine similarity matches.

3. CAMPUS LOCATION / LANDMARK

Campus Landmark

CSE Block Canteen

4. PHOTO (OPTIONAL) Stored client-side as base64

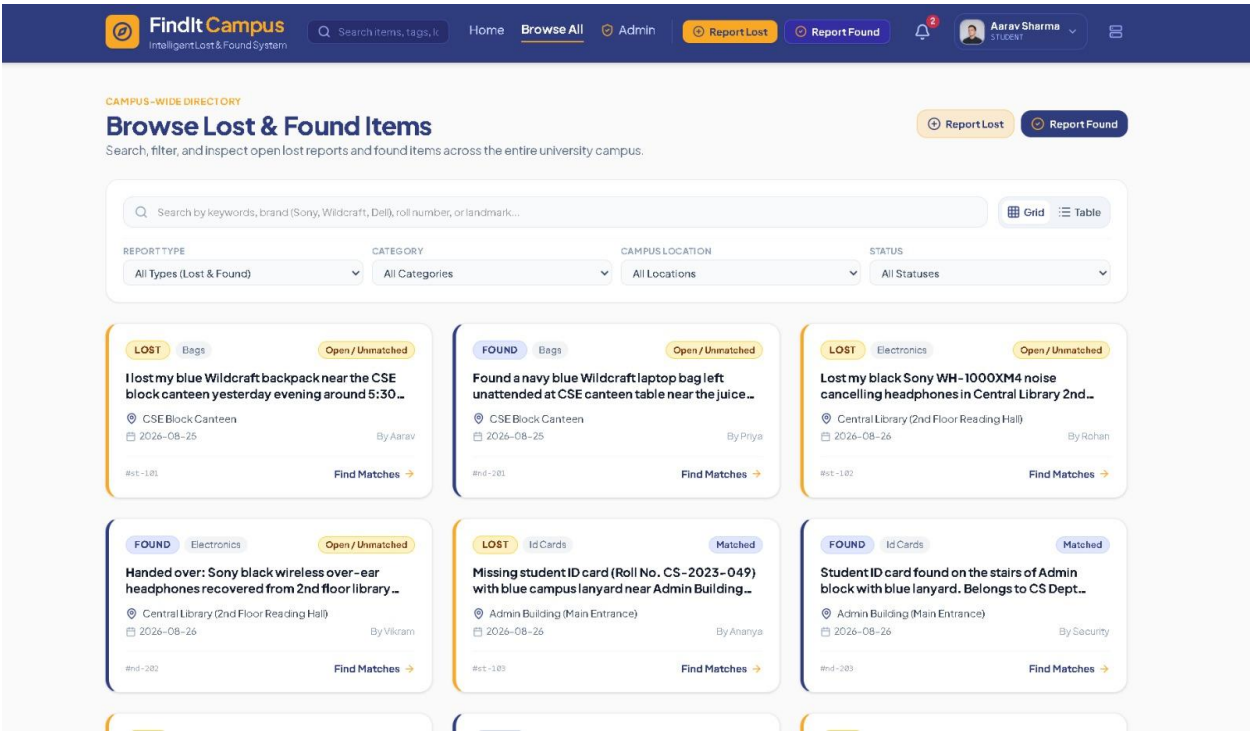
Drag and drop item image here, or browse file
PNG, JPG, WebP up to 5MB

Submitting triggers automatic real-time vector matching against opposite reports.

Submit Found Item Report

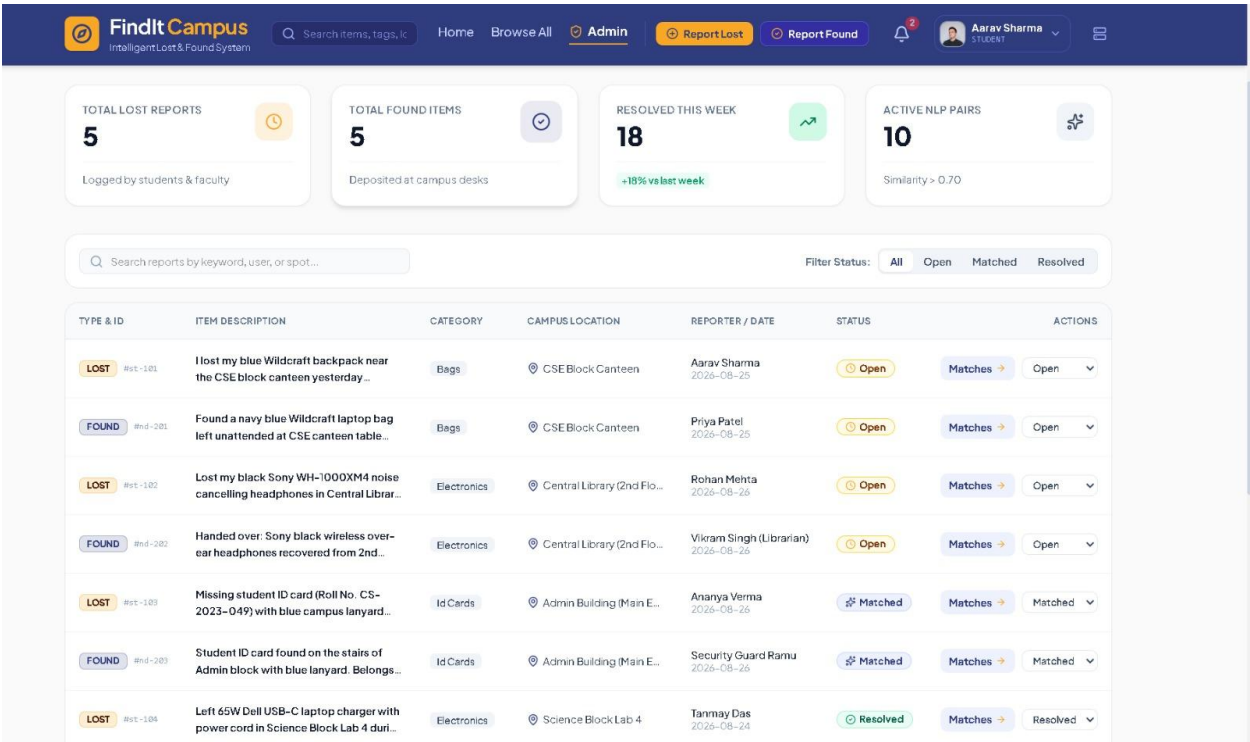
Report Found form mirroring the lost-item workflow for recovered items.

Figure A3 – Browse Lost & Found Items Directory



Campus-wide directory with search, category, location, and status filters.

Figure A4 – Administrator Dashboard



Admin console showing report volume, matching activity, weekly resolution indicators, and status controls.

Appendix B — NLP Pipeline Summary

The implemented conceptual pipeline is: User Report → Text Preprocessing → NER → HMM-based POS Tagging → TF-IDF + N-grams → Feature Vector → Cosine Similarity + Location Matching → Thresholding and Ranking → Possible Match → Human Verification → Matched / Resolved.

Table B1 – NLP Pipeline Summary

Module	Technique	Primary Output
Module 1	NER	Extract item, location, time, colour, brand and related entities
Module 2	HMM-based POS	Identify grammatical word categories and structure
Module 3	TF-IDF + N-grams	Create comparable numerical/text features
Module 4	Cosine Similarity	Score, threshold, and rank candidate reports

Appendix C – Source Code / Code Repository Information

This appendix provides information about the source code and implementation resources used for the Campus-Aware Intelligent Lost and Found Management System Using Natural Language Processing (NLP).

The implementation consists of a modular NLP pipeline integrated with the FindIt Campus reporting and matching workflow. The major software components include:

Component	Implementation / Purpose
User Interface	Lost-item and found-item reporting forms
Backend	Report validation, storage and matching workflow
NER Module	Extraction of item, location, time, colour and brand information
POS Module	HMM-based Part-of-Speech tagging
Feature Extraction	TF-IDF with unigram, bigram and trigram features

Similarity Module	Cosine similarity and location-aware comparison
Matching Module	Threshold-based candidate ranking
Administration	Match verification and report-status management

Appendix D – Datasheets / Software Requirements

Since FindIt Campus is primarily a software-based NLP application, conventional hardware datasheets are not applicable to the core implementation. The following software requirements are relevant to the project.

Software / Technology	Purpose
Python	NLP processing and implementation
NLTK	Natural language processing and HMM-based POS tagging
Scikit-learn	TF-IDF feature extraction and vector processing
Web Application Framework	User reporting and administrative interface
Database	Storage of lost/found reports and match information
HTML/CSS/JavaScript	User-interface development
Development Environment	Code development, testing and debugging

Appendix E – Experimental / Raw Data

This appendix contains the representative experimental data used for testing the FindIt Campus matching system.

The evaluation included lost and found reports from Bags & Backpacks, Electronics, and ID Cards, together with an unmatched charger case. Three known lost/found pairs were used to verify whether the correct counterpart was ranked first.

Case	Category	Experimental Purpose	Result
Backpack	Bags & Backpacks	Test matching with vocabulary variation	Correct counterpart ranked first

Headphones	Electronics	Test partial description overlap	Correct counterpart ranked first
Student ID Card	ID Cards	Test strong attribute similarity	Correct counterpart ranked first
65W Dell USB-C Charger	Electronics	Test unmatched condition	Remained Open / Unmatched

The test environment included five lost reports and five found submissions covering locations such as CSE Block Canteen, Central Library, Admin Building and Science Block Lab. These values represent the project test snapshot and are not intended as a statistically representative campus-wide dataset.

The actual raw dataset, if maintained separately, should be submitted with the project materials where institutional and privacy requirements permit.

Appendix F – Ethics / Institutional Approval

The FindIt Campus system handles lost-and-found descriptions and may contain user-provided information such as reporter details and item descriptions. Ethical considerations were therefore incorporated into the system design.

The project follows the principles of:

1. Data minimization.
2. Collection of only necessary information.
3. Restricted access to administrative information.
4. Human verification before final ownership resolution.
5. Transparency regarding automated NLP processing.
6. Avoidance of forced matches when similarity is insufficient.
7. Protection against incorrect assignment of property.

A false-positive match could result in an item being given to the wrong person. Therefore, the NLP system recommends possible matches rather than making the final ownership decision.

Institutional Approval Status:

The available project report **does** not document a separate institutional ethics approval or approval number. If formal institutional approval was required or obtained, the relevant approval letter/certificate should be attached here. Otherwise, the ethical and responsible data-handling procedures documented in Chapter 3 should be retained as the project's ethics evidence.

Appendix E – Individual Contribution Evidence

The project was completed as a team effort, with each member taking primary responsibility for one core NLP module while contributing to integration, testing, documentation and presentation.

Individual Contribution Evidence

Student	Primary Module	Technical Contribution	Suggested Evidence
A. Syed Thoufiq	Module 1 – Campus Context & Item Information Extraction	NER pipeline; item, location and date/time extraction	NER source code, entity-extraction outputs, test screenshots
S. Mohammad Abrar	Module 2 – Syntactic Structure Analysis	HMM-based POS tagging and grammatical category identification	POS-tagging code, sample outputs, testing records
B. Praneeth	Module 3 – Linguistic Feature Extraction	TF-IDF, unigram/bigram/trigram extraction and feature-vector construction	Feature-extraction code, vector outputs, analysis results
K. Subrahmanya Venkata Sai Manikanta	Module 4 – Lost-and-Found Similarity Matching	Cosine similarity, location-aware comparison, ranking and thresholding	Similarity code, score outputs, ranking results and test cases